module 5

November 5, 2026

0.1 Module 5

In this assignment, you are going to work on Histograms and Scatterplots.

We have preprocessed the data as "df" for you.

Follow the instructions and finish the rest part.

[1]:
```
%%capture
################################################################
### EXECUTE THIS CELL BEFORE YOU TO TEST YOUR SOLUTIONS ###
################################################################
import imp, os, sys
sol = imp.load_compiled("solutions", "./solutions.py")
sol.get_solutions("imdb.xlsx")
from nose.tools import assert_equal
from pandas.util.testing import assert_frame_equal, assert_series_equal
import warnings
warnings.filterwarnings('ignore', category=UserWarning, module='openpyxl')
```

[16]:
```
# Loading the data
import pandas as pd
import numpy as np

xls = pd.ExcelFile('imdb.xlsx')
df = xls.parse('imdb')
df_directors = xls.parse('directors')
df_countries = xls.parse('countries')

df = pd.merge(left=df, right=df_countries,
              how='inner', left_on='country_id',
              right_on='id')

df = pd.merge(left=df, right=df_directors,
              how='inner', left_on='director_id',
              right_on='id')

print("Finished.")
```

Finished.

[28]:
```
"""Q1:
Is how much a movie makes indicative of how good it is?
Make a simple scatter plot comparing gross to imdb_score for movies during or␣
 ↪after 2000 (title_year >= 2000) and before 2000 (title_year < 2000).
It may be useful to scale the x axis demarking gross. (Hint: Divide the gross␣
 ↪amount by 1,000,000.)
Remember to put a legend indicating which color corresponds to which years.
What is your verdict?

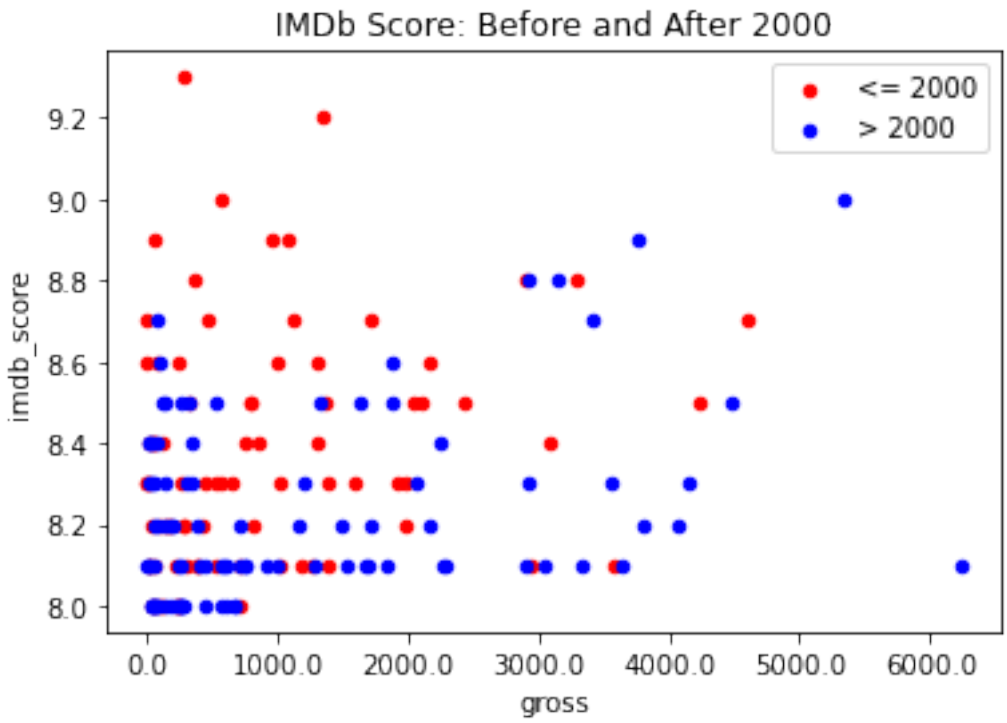
Save your plot in a variable called plt1, and your dataframes in variables␣
 ↪called df_after_2000 and df_before_2000
"""

import matplotlib.pyplot as plt1

# your code here
df_before_2000 = df[df.title_year < 2000]
df_after_2000 = df[df.title_year >= 2000]

ax = df_before_2000.plot.scatter(
    x='gross', y='imdb_score', color='red', label='<= 2000'
)

df_after_2000.plot.scatter(
    x='gross', y='imdb_score', color='blue', label='> 2000', ax=ax
)
ax.set_xticklabels(ax.get_xticks() / 100000)
ax.set_title('IMDb Score: Before and After 2000')
ax.legend()
plt1.show()
```



[29]:
```
assert_frame_equal(df_before_2000, sol.df_before_2000)
assert_frame_equal(df_after_2000, sol.df_after_2000)
np.testing.assert_array_equal(plt1, sol.plt1)
print("Success!")
```

```
Success!
```

[37]:
```
"""Q2:
Using numpy and pyplot, make an overlapping histogram that shows the score␣
 ↪distribution vs. count of R-Rated movies and PG-13 ones.
Describe your plot.

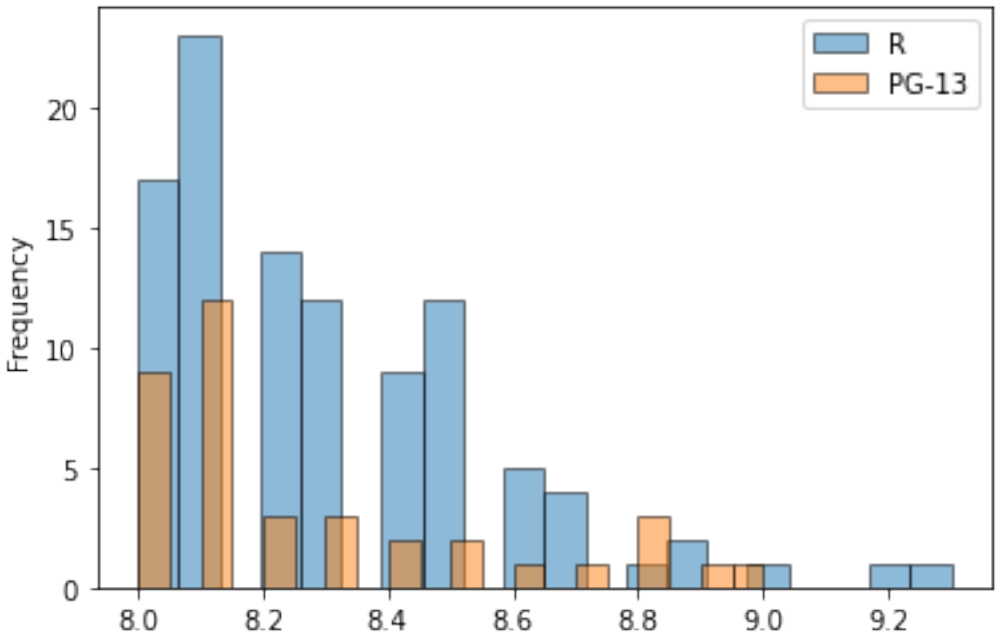
Save your plot in a variable called plt2, and your dataframes in variables␣
 ↪called df_R and df_PG13
"""

import matplotlib.pyplot as plt2

# your code here
df_R = df[df.content_rating == 'R']
df_PG13 = df[df.content_rating == 'PG-13']
ax = df_R['imdb_score'].plot.hist(bins=20,  alpha=0.5, edgecolor='black',␣
 ↪label='R')
```

```
ax = df_PG13['imdb_score'].plot.hist(bins=20,  alpha=0.5, edgecolor='black',␣
 ↪ax=ax, label='PG-13')
ax.legend()

plt2.show()
```



[38]:
```
#df_R.head()
```

[39]:
```
assert_frame_equal(df_R, sol.df_R)
assert_frame_equal(df_PG13, sol.df_PG13)
np.testing.assert_array_equal(plt2, sol.plt2)
print("Success!")
```

```
Success!
```

[ ]:
```

```